MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11746254
CONTROL PANEL-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of Specification MIL-C-48411 comprise the complete requirements for Control Panel-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-G-48411 | Circuit Card Assemblies, General Specification for |

DRAWINGS

| | |
|---|---|
| 11746254 | Circuit Card Assembly, Control Panel-1 |

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item Definition.- This assembly is one (1) of two (2) single ended printed circuit boards consisting of gates, flip-flops, and inverters necessary to provide the logic circuitry to light the various lamps located on the front panel of the Simulated Target Generator.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746254.



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3.2 Characteristics.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power supply.- The requirements of 3.2 shall be met with the following power supply voltages.

a. +5 VDC
b. +28 VDC

3.2.1.2 Clock.- A 15 Hz clock input shall be capable of being reduced and inverted when applied to Pin 25.

3.2.1.3 Voltage Divider.- The outputs at Pin 3 & 5 shall be 2.5 ± 0.05 VDC.

3.2.1.4 Logic Light Switches.- The outputs at Pins 9, 12, 18, 20, 23, and 27 shall be capable of energizing and de-energizing MS 25237-327 bulbs or its equivalent.

3.2.1.5 Clock Pulse.- The clock pulse shall be capable of being logically switched on and off and shall have a pulse duration of 33 milliseconds.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-G-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11746254 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-G-48411, Drawing 11746254 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-G-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-G-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-G-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.



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4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

4.2.1 Test equipment.- (see Figure 1)

4.2.1.1 Power Supply (PS1).- The power supply shall be 10 ± 0.5 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Power Supply (PS2).- The power supply shall be 40 ± 2.0 VDC, 200 mA or greater and a ripple of less than 20 mV RMS.

4.2.1.3 Oscilloscope.- Rise time of scope and associated measuring probe of 50 nanoseconds or less, sweep period of 50 nanoseconds per centimeter or greater, input resistance greater than 8 megohms, input capacitance less than 25 picafarads.

4.2.1.4 Signal generator.- The generator shall be capable of a 15 Hz ± 10%, 0 to 5 VDC, and 10 nanoseconds rise and fall time square wave output.

4.2.1.5 Dummy Loads.- The dummy loads shall be one 3-input positive "nand" gates found in a 54L10, 5410 or equivalent, and miniture bulbs MS25237-327 or their equivalent.

4.2.1.6 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power supply.- All the requirements shall be met with the following power supply voltages.

a. +5 ± 0.05 VDC
b. +28 ± 2.0 VDC

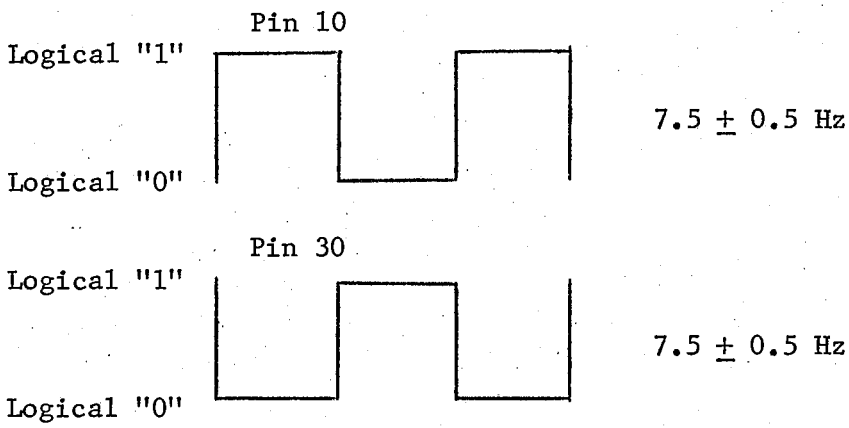


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4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

4.2.2.3 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.4 Clock.- With a 15 ± 0.5 Hz, 5 volt square wave pulse applied to Pin 25, the outputs on Pins 10 and 30,as shown below, shall be 7.5 ± 0.5 Hz. The output on Pin 10 shall be in phase with the input and for Pin 30 shall be 180 degrees out of phase.




4.2.2.5 Voltage Divider.- When 5.00 VDC is applied to the circuit board the outputs on Pins 3 and 5 are 2.5 ± 0.05 VDC.

4.2.2.6 Logic Light Switches.- With the circuit board connected shown in Figure 1 the inputs as listed in Tables below shall yeild the corresponding outputs.

TABLE I

| Input (Logic/VDC) | | Output |
|---|---|---|
| Pin 1 | Pin 2 | Pin 9 (L1) |
| Logic "0" | Logic "0" | Off |
| Logic "1" | Logic "0" | On |
| +28 VDC | Logic "0" | On |

TABLE II

| Input (Logic/VDC) | | Instructions | Output (Logic/Lamp) | | |
|---|---|---|---|---|---|
| Pin 33 | Pin 35 | | Pin 38 | Pin 40 | Pin 12(L2) |
| +28VDC | Logic 0 | Reset "0" | Logic 0 | Logic 1 | Off |
| +28VDC | Logic 0 | Reset "1", Reset "1" | Logic 1 | Logic 0 | On |
| +28VDC | +28VDC | Reset "0", Reset "1", Reset "1" | | | Blinks |



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TABLE III

| Input (Logic/VDC) | | Output (Lamp) |
|---|---|---|
| Pin 33 | Pin 35 | Pin 18(L3) |
| Logic "0" | Logic "0" | Off |
| +28VDC | Logic "0" | On |
| +28VDC | +28VDC | Off |
| Logic "0" | +28VDC | Off |

TABLE IV

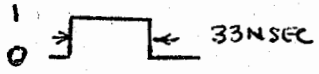
| Input (Logic/VDC) | | Instructions | Output | | |
|---|---|---|---|---|---|
| Pin 33 | Pin 35 | | Pin 28 | Pin 29 | Pin 20(L4) |
| +28VDC | Logic "0" | Reset "0" | Logic "1" | Logic "0" | Off |
| +28VDC | Logic "0" | Reset "1" | Logic "0" | Logic "1" | On |
| +28VDC | +28VDC | Reset "0", Reset "1" | | | Blinks |

TABLE V

| Input (Logic) | | Instructions | Output (Lamp) |
|---|---|---|---|
| Pin 24 | Pin 32 | | Pin 23(L5) |
| Logic "0" | Logic "0" | Reset "0" | Off |
| Logic "1" | Logic "0" | | On |
| Logic "1" | Logic "1" | | Blinks |
| Logic "0" | Logic "1" | Reset "0" | Off |

4.2.2.7 Clock Pulse.- With the circuit board connected as shown in Figure I, the inputs as listed in the Table below shall yeild the corresponding results.

TABLE VI

| Input(Logic/VDC) | | Output (Logic/Wave Form) |
|---|---|---|
| Pin 33 | Pin 34 | Pin 36 |
| Logic "0" | Logic "0" | Logic "0" |
| Logic "0" | +28VDC | Logic "0" |
| +28VDC | +28VDC |  |
| +28VDC | Logic "0" | Logic "0" |





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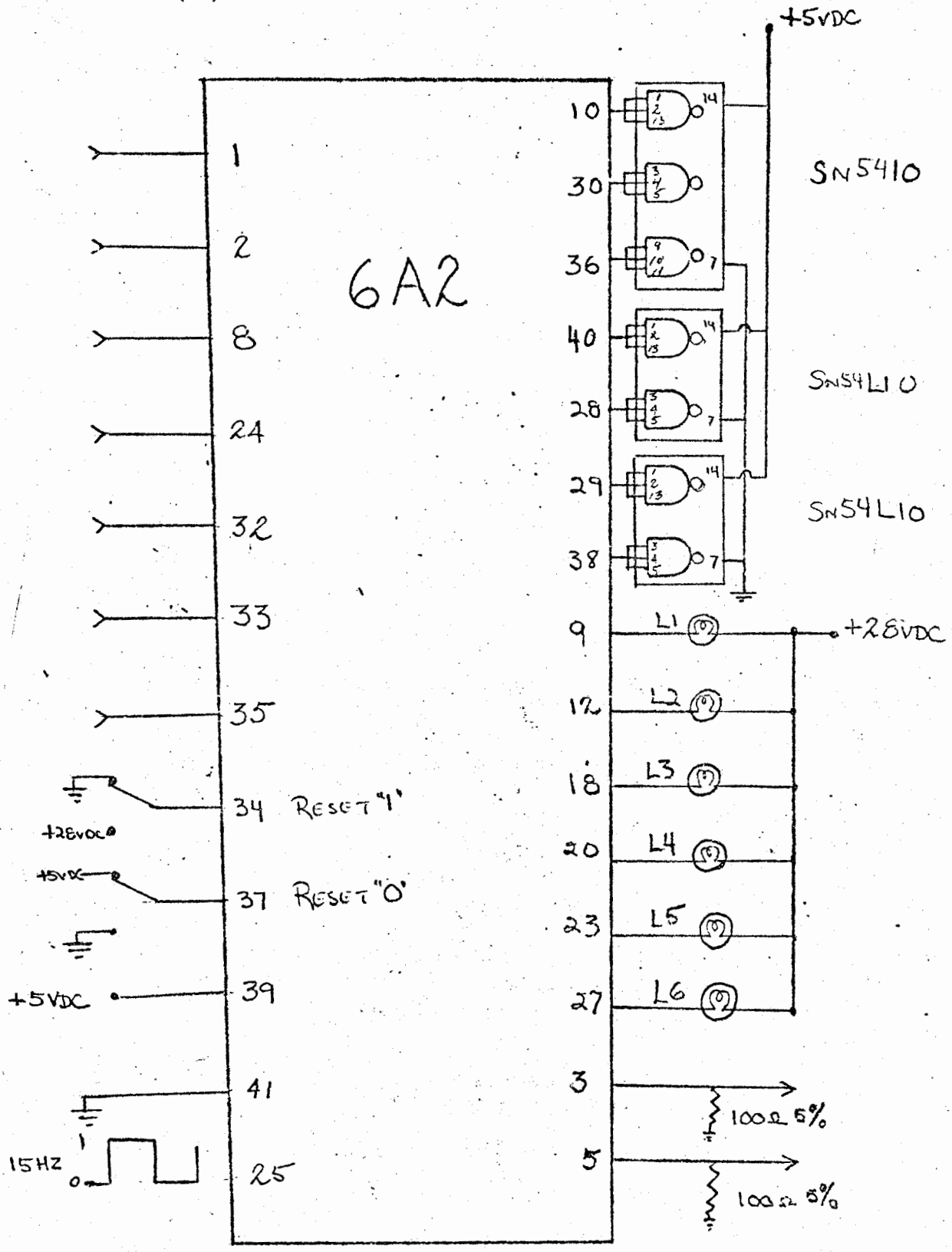



FIGURE I



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TABLE VII

| Input (Logic/VDC) | | Instructions | Output | | |
|---|---|---|---|---|---|
| Pin 33 | Pin 35 | | Pin 38 | Pin 40 | L6 |
| +28VDC | Logic "0" | Reset "0" | Logic "0" | Logic "1" | Off |
| +28VDC | Logic "0" | Reset "1", Reset "1" | Logic "1" | Logic "0" | On |
| +28VDC | +28VDC | Reset "0", Reset "1", Reset "1" | | | Blinks |

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746254.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applicable to this specification.

Custodian:

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Preparing Activity:

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